
Simultaneous Speech-to-Speech Translation Without Aligned Data

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Abstract

Simultaneous speech translation requires translating source speech into a target language in real-time while handling non-monotonic word dependencies. Traditional approaches rely on supervised training with word-level aligned data, which is difficult to collect at scale and thus depends on synthetic alignments using language-specific heuristics that are suboptimal. We propose Hibiki-Zero, which eliminates the need for word-level alignments entirely. This fundamentally simplifies the training pipeline and enables seamless scaling to diverse languages with varying grammatical structures, removing the bottleneck of designing language-specific alignment heuristics. We first train on sentence-level aligned data to learn speech translation at high latency, then apply a novel reinforcement learning strategy using GRPO to optimize latency while preserving translation quality. Hibiki-Zero achieves state-of-the-art performance in translation accuracy, latency, voice transfer, and naturalness across four X-to-English tasks. Moreover, we demonstrate that our model can be adapted to support a new input language with less than 1000h of speech. We provide examples, model weights, inference code¹ and we release a benchmark containing 45h of multilingual data for speech translation evaluation.²

1. Introduction

We introduce Hibiki-Zero, a system for simultaneous and expressive speech-to-speech (S2ST) and speech-to-text (S2TT) translation that does not require aligned data for training. Unlike offline speech translation systems that access the full source utterance before translating, simultaneous translation

must produce output incrementally while maintaining both translation accuracy and speech naturalness. This requires learning a fine-grained translation policy that determines when to listen and when to speak. The most straightforward approach to learning such a policy is through supervised training on aligned data. However, human interpretation data with word-level alignments is virtually non-existent, forcing state-of-the-art systems to rely on synthetic data with automatic alignments (Labiausse et al., 2025). These automatic alignments are inherently limited, as they depend on hand-crafted heuristics rather than being learned from data.

Hibiki-Zero is a decoder-only model that synchronously receives source speech and generates translated speech leveraging a multistream architecture originally introduced by Défossez et al. (2024). Unlike Hibiki (Labiausse et al., 2025), Hibiki-Zero is not trained with supervised learning on synthetic interpretation data but rather casts joint optimization of translation quality and latency as a reinforcement learning (RL) problem. While we still require a base model before the RL phase, it is trained using sentence-level aligned data which can be more easily constructed independently of the language compared to word-level aligned data. During RL, we exploit the sentence-level aspect of our data to design a simple reward system based on BLEU score (Papineni et al., 2002) only. To achieve this, we compute rewards at multiple intermediate instants during the translation of an input speech utterance by leveraging the simultaneous text translation also produced by our model. Using these *process rewards*, we obtain fine-grained local *advantages* across multiple translations from the same input. We then adapt GRPO (Shao et al., 2024) to our multistream architecture, using these advantages to optimize the model.

In a multilingual-to-English translation task, Hibiki-Zero outperforms previous state-of-the-art work in translation quality, latency, speaker identity preservation, and speech naturalness. We also retain all the benefits of multistream modeling such as batching and real-time inference on GPU while removing the necessity to build interpretation-like training data thus considerably simplifying the development of such models. We even demonstrate that Hibiki-Zero can adapt to a new input language with less than 1000h of training data marking an important step to make high quality speech translation (ST) available in more languages.

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Proceedings of the 43rd International Conference on Machine Learning, Seoul, South Korea. PMLR 306, 2026. Copyright 2026 by the author(s).

¹github.com/kyutai-labs/hibiki-zero

²huggingface.co/collections/kyutai/hibiki-zero

Conflict of interest disclosure. Authors N.Z. and A.D. are employed by Gradium, which develops one of the three TTS models used to create the long-form evaluation dataset.

2. Related Work

2.1. Simultaneous end-to-end speech translation

While speech translation was initially performed using cascaded systems combining automatic speech recognition (ASR), machine translation (MT) and text-to-speech synthesis (TTS) (Baldridge, 2004; Nakamura et al., 2006), it recently evolved in fully end-to-end systems (Jia et al., 2019; Lee et al., 2022a; Jia et al., 2022; Rubenstein et al., 2023) reducing error propagation and enabling transfer of non-linguistic information such as the speaker voice identity or prosody to the generated speech. At first trained with auxiliary text of phoneme translation tasks (Jia et al., 2022; Zhang et al., 2024a), most recent works (Barrault et al., 2023; Labiausse et al., 2025; Cheng et al., 2025; Misiunas & Ablavatski, 2025) train directly on simultaneous S2TT and S2ST tasks so they can use the predicted text translation as a scaffolding for speech generation at inference time. Among direct ST training methods, those who achieve better speech naturalness are duplex audio systems that require to build a simultaneous ST dataset. They either rely on a synthetic data generation pipeline which includes a fine word-level text-to-translation alignment method (Labiausse et al., 2025; Misiunas & Ablavatski, 2025) or use a text LLM to split text into semantic chunks (a few words) that are individually translated thus providing chunk-level translation alignment (Cheng et al., 2025) before collecting human-annotated interpretation data for finetuning purposes. Hibiki-Zero removes most of the complexity from synthetic data generation as it only requires sentence-level translation alignment easily obtained from punctuation. Thanks to an efficient RL process, it is then possible to reduce the translation latency of the model so it achieves state-of-the-art quality/latency trade-off in multiple input languages.

2.2. Self-improvement of real-time translation systems

RL methods to improve simultaneous translation systems were first explored in the context of text translation. Some works used preference-based approaches (Yu et al., 2025) with preferences established in the context of simultaneous ST by prompting a text LLM while others applied online reinforcement procedures (Yu et al., 2025; Xu et al., 2025) with sequence-level rewards as a combination of translation quality and latency metrics. Because they lack sub-sentence granularity in their preference or reward signals, it is difficult for these methods to find an appropriate balance between translation quality and latency during the RL process. More recently, Seed LiveInterpret 2.0 (Cheng et al., 2025) applied PPO (Schulman et al., 2017) with a combination of inter-

mediate evaluations of the generated sequences (*process rewards*) and overall evaluation of the translation (*outcome rewards*). Starting from a base supervised ST model trained with chunk-level alignment and finetuned on high-quality human interpretation data, they managed to strictly improve the quality/latency trade-off through RL. However, they combine multiple quality and latency measures to compute their reward thus introducing hyperparameters that are difficult to tune in practice. Because of complex interactions between the different reward components, they encountered training instability, reward hacking and had to rely on two different stages of RL training, using only outcome rewards at first before adding process rewards. On the other hand, Hibiki-Zero uses a single and straightforward reward system that unifies the translation quality and latency criteria into a single reward derived from BLEU (Papineni et al., 2002) scores only. Reinforcement learning is performed using GRPO (Shao et al., 2024) without KL regularization as previously done by Rastogi et al. (2025) to reduce memory requirements during training. Most importantly, it does not rely on any human interpretation or annotated data to finetune the model before reinforcement. On multilingual simultaneous ST tasks, Hibiki-Zero achieves state-of-the-art translation quality, translation latency, naturalness and speaker identity preservation. Hibiki-Zero is even able to adapt to a new input language after a light finetuning.

3. Method

We consider an utterance in a source language represented as a monophonic waveform $X \in \mathbb{R}^{f_s \cdot d}$, sampled at a frame rate $f_s = 24$ kHz, of duration d . Similarly, its translation is given in a target language, denoted $Y \in \mathbb{R}^{f_s \cdot d}$. We assume X is padded to ensure both have the same duration. Our objective is to model $\mathbb{P}[Y|X]$. Contrary to Labiausse et al. (2025), we do not constrain the modeling of Y knowing X to be entirely causal in our training data. Thanks to the diversity of causality and latency arrangements in the dataset, it is still possible to learn a base translation model. Its behavior is then adjusted by an online reinforcement learning strategy that rewards correct and simultaneous translations.

3.1. Modeling

We build on the framework introduced by Défossez et al. (2024) for the joint modeling of multiple sequences of tokens and used by Labiausse et al. (2025) to perform simultaneous S2TT and S2ST with high fidelity.

3.1.1. NEURAL AUDIO CODEC

We use the pre-trained causal and streaming Mimi codec (Défossez et al., 2024) to encode X and Y into low framerate sequences of discrete tokens. Mimi consists of an encoder and decoder from and to the waveform domain,

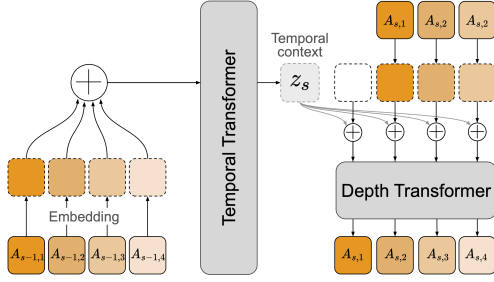


Figure 1. **Architecture of the RQ-Transformer.** Figure adapted from Défossez et al. (2024).

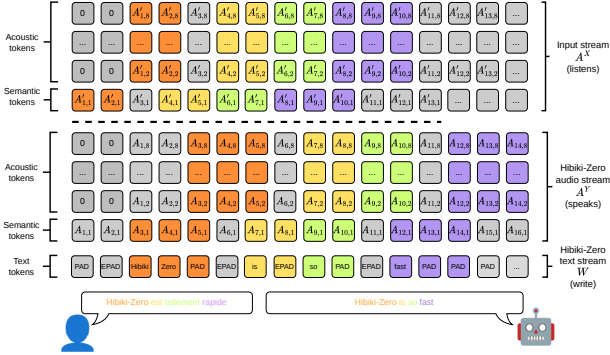


Figure 2. **Joint sequence modeling.** From the input stream, Hibiki-Zero predicts its Inner Monologue text stream, semantic and acoustic tokens. Figure adapted from Labiausse et al. (2025).

and of an information bottleneck using Residual Vector Quantization (RVQ) (Zeghidour et al., 2022).

For language modeling, we are interested in the discrete indices of codebook entries which Mimi latents are projected to. We denote those $(A_{t,q}) \in \{1, \dots, N_a\}^{f_r \cdot d \times Q}$ where $f_r = 12.5$ Hz is the codec framerate, Q is the number of audio residual quantization levels varying up to 32 and N_a the codebooks size. Following Zhang et al. (2024b); Défossez et al. (2024), the output of the first quantization level is trained to replicate semantic information obtained from a WavLM self-supervised audio model (Chen et al., 2022). We refer to $A_{t,1}$ as *semantic* tokens, and $A_{t,q \geq 2}$ as *acoustic* tokens with the latter arranged in a coarse to fine manner. When encoding audio, we keep only the first $Q = 16$ acoustic levels as they are sufficient to ensure high quality speech reconstruction through Mimi’s decoder.

3.1.2. JOINT MODELING OF DISCRETE AUDIO TOKENS

Following Yang et al. (2023); Labiausse et al. (2025), we leverage a RQ-Transformer (Lee et al., 2022b) as shown in Figure 1 to model $(A_{t,q})$ both over the time t and quantizer q axes as audio streams cannot be reasonably merged into a single discrete sequence. It consists in a large *Temporal* Transformer (Vaswani et al., 2017) of latent dimension D , operating at the same framerate f_r as the codec, and being

fed all the tokens generated so far, e.g. for all $t \leq f_r \cdot d$,

$$Z_t = \text{Temp}(A_0, \dots, A_{t-1}) \in \mathbb{R}^D. \quad (1)$$

A_0 is defined as a deterministic token indicating the start of the generation. Then, a smaller scale *Depth* Transformer models auto-regressively the tokens $A_{t,1}, \dots, A_{t,Q}$ over the quantizer axis, e.g. for all $t \leq f_r \cdot d$ and $q \leq Q$,

$$l_{t,q} = \text{Dep}(Z_t, A_{t,0}, \dots, A_{t,q-1}) \in \mathbb{R}^{N_a}, \quad (2)$$

with $A_{t,0}$ also a special token, and with the goal of having,

$$\text{softmax}(l_{t,q}) \approx \mathbb{P}[A_{t,q} | A_0, \dots, A_{t-1}, A_{t,0}, \dots, A_{t,q-1}]$$

Following (Copet et al., 2023; Défossez et al., 2024), we introduce an acoustic delay shifting acoustic tokens of 2 time steps in the future compared to the semantic stream. The streams are realigned before decoding the audio with the codec. As this delay is always applied, we don’t introduce new notations for readability and refer to $(A_{t,q})$ directly.

3.1.3. TRANSLATION AS MULTISTREAM MODELING

Using the RQ-Transformer given by Eq. (1) and (2) to jointly model multiple discrete streams of tokens, we can perform the task of joint simultaneous S2TT and S2ST as illustrated in Figure 2. Following (Défossez et al., 2024), we use an *Inner Monologue* to improve stability of the generated audio content. We introduce a stream of padded text tokens $(W_t) \in \{1, \dots, N_W\}^{f_r \cdot d}$ whose content is the word-level aligned text transcription of the audio modeled in A^Y . This text stream is concatenated with the audio tokens A^Y along the q -axis such that it comes before the semantic level. The resulting multistream of target tokens are themselves concatenated with the source tokens A^X along the q -axis. At inference time, predictions of tokens A^X are skipped and actual tokens of the input audio are forced instead.

3.1.4. ARCHITECTURAL DETAILS

At time-step t , tokens from the previous step, e.g. A_{t-1}^X , A_{t-1}^Y , and W_{t-1} , are fed into dedicated embedding tables and contributions are summed with a BOS token used for the first time step $t = 1$. The RQ-Transformer uses standard Transformer layers (Vaswani et al., 2017), with gated SiLU activation (Shazeer, 2020; Hendrycks & Gimpel, 2016). A linear layer maps output Z_t of the *Temporal* Transformer to logits for the text token W_t . The *Depth* Transformer then operates for Q steps to estimate the logits for the output stream and for Q additional steps for the input stream. Each depth step q takes as input Z_t summed with a learned embedding of the previous audio token $A_{t,q-1}$, or W_t for $q = 1$. We provide architectural hyper-parameters in Section 4.1.

3.2. Coarse alignment of speech translation data

We have assumed training pairs (X, Y) to not be entirely causal at the interpretation level. We now detail the specific method used to build such coarse translation alignments.

3.2.1. SENTENCE-LEVEL ALIGNMENT

We start from an unaligned speech translation pair (X, Y) which only verifies a sentence mapping constraint meaning that both X and Y contain the same number of sentences and such that the i^{th} sentence in Y is a translation of the i^{th} sentence in X . Inspired by Labiausse et al. (2025), we rely on the insertion of artificial silence in Y to delay its content with respect to X . For each sentence of index i , we introduce silence in Y to shift its i^{th} sentence by an amount δ_i after the start of the i^{th} sentence in X where $\delta_i \sim \mathcal{U}(0, \delta \times d_i)$ is sampled independently for each sentence, d_i is the duration of the i^{th} sentence in X and $\delta \in [0, 1]$ is an hyperparameter. Then, using punctuation characters such as commas or colons in a precomputed transcript of Y , we insert silences whose durations follow $\mathcal{U}(0, \mu)$ at the corresponding instants in Y with μ being a hyperparameter.

3.2.2. NATURAL PAUSES TTS

Using the method described in Section 3.2.1, we might break the natural flow of speech by inserting silence on punctuations which is also subject to imprecisions of the transcript timestamps. Following Zeghidour et al. (2025), we train a TTS with synced audio and text streams as output, providing a control on the emission timestamp of each word to synthesize. Moreover, we train the TTS to perform voice transfer from a short audio conditioning of maximum 10 seconds. We can then generate an audio Y^+ using the original transcript of Y and naturally insert the pauses described in 3.2.1 while conditioned on the speaker from X . This results in new training pairs (X, Y^+) where targets contain smoother transitions between speech and silences than Y .

3.3. Translation policy reinforcement

Assuming that we dispose of a simultaneous translation model as presented in Section 3.1, we now introduce a reinforcement learning procedure using process rewards based on BLEU scores to improve the translation policy of the model as illustrated in Figure 3. We adapt GRPO from Shao et al. (2024) to be our RL algorithm. We denote by π_θ the translation model to optimize and $\pi_{\theta_{old}}$ an older version of it acting as a regularizer. Given an input speech utterance X with a known sentence-level text translation y , we use $\pi_{\theta_{old}}$ to generate G different speech translations $(Y_i)_{1 \leq i \leq G}$, each of duration $T \times f_r$ seconds where f_r is the model frame rate and T a fixed number of frames.

3.3.1. PROCESS REWARDS

Let n be the number of sentences in X and $(t_i)_{0 \leq i \leq n}$ the frame indexes such that the sentence of index i in X starts at frame t_i and ends at frame t_{i+1} . We introduce $S(t)$ as the sentence index at frame $t \geq t_0$ in X i.e. $S(t) = i$ for $t_i \leq t \leq t_{i+1}$ and $S(t) = n - 1$ for $t > t_n$. For a frame index $t \leq T$, we denote y_t as the text concatenation of translated input sentences until the one of index $S(t)$ included. Given a generation i , we define $\hat{y}_t^{(i)}$ as the partial text transcript until frame t given by the model's output text stream. We now introduce the hyperparameter $\alpha \in [0, 1]$ and define the process reward for generation i at frame t as:

$$r_t^{(i)} = (1 - \alpha)\text{BLEU}(\hat{y}_t^{(i)}, y_t) + \alpha\text{BLEU}(\hat{y}_T^{(i)}, y_T) \quad (3)$$

3.3.2. OPTIMIZATION OBJECTIVE

Using the modeling of X and $(Y_i)_{1 \leq i \leq G}$ as tokens A^X and $A^{Y_1}, \dots, A^{Y_G}$, we define the probability ratios between π_θ and $\pi_{\theta_{old}}$ for each output i , codebook index $q \leq Q$ and frame index $t \leq T$ as:

$$p_{q,t}^{(i)} = \frac{\pi_\theta(A_{q,t}^{Y_i} | A_{\leq t}^X, A_{< t}^{Y_i}, A_{< q,t}^{Y_i})}{\pi_{\theta_{old}}(A_{q,t}^{Y_i} | A_{\leq t}^X, A_{< t}^{Y_i}, A_{< q,t}^{Y_i})} \quad (4)$$

Given a set of frame indexes $t'_1 < t'_2 < \dots < t'_s$, we compute process rewards as defined in Section 3.3.1 for each output, namely $r_{t'_j}^{(i)}$ for $i \leq G$ and $j \leq s$. We then normalize rewards per frame index across group elements:

$$\bar{r}_{t'_j}^{(i)} = \frac{r_{t'_j}^{(i)} - \text{mean}_{k \leq G}[r_{t'_j}^{(k)}]}{\text{std}_{k \leq G}[r_{t'_j}^{(k)}]} \quad (5)$$

In practice, early experiments showed that using a regular frame indexes pattern along the input speech content performed better. Thus we introduce $n_w \in \mathbb{N}^*$ and use the end timestamp of every n_w words in the input to set $(t'_j)_{j \leq s}$.

Then, we introduce the advantage of an output at step t as the sum of normalized rewards from the following steps:

$$R_t^{(i)} = \sum_{t'_j > t} \bar{r}_{t'_j}^{(i)} \quad (6)$$

We compute the per-codebook objectives $L_q^{(i)}$ using the standard clipping function between $1 - \epsilon$ and $1 + \epsilon$ as:

$$L_q^{(i)} = \sum_{t=1}^T \min(p_{q,t}^{(i)} R_t^{(i)}, \text{clip}_\epsilon(p_{q,t}^{(i)} R_t^{(i)})) \quad (7)$$

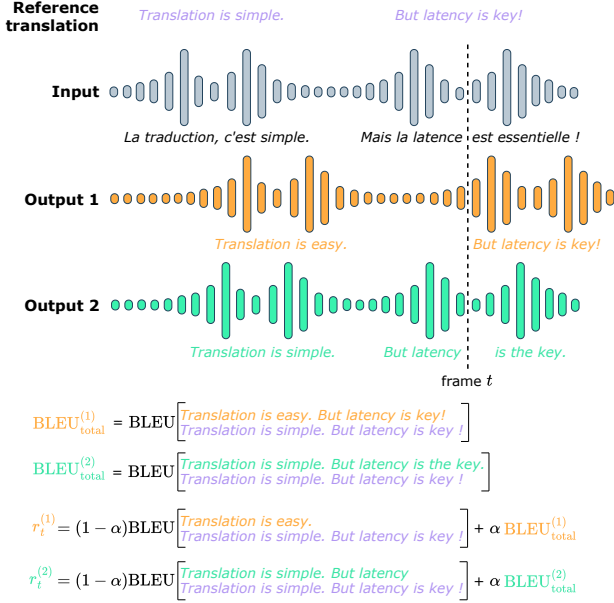


Figure 3. **Process rewards method based on BLEU score.** We introduce intermediate BLEU score computed on the text output of the model before a given frame t and using the ground-truth translation of the corresponding input sentences processed so far. We combine it with the total output BLEU score using $\alpha \in [0, 1]$.

In the end, we seek to maximize the following objective with fixed weights c_q for each depth $q \leq Q$:

$$\mathbb{E}_{\substack{X \sim \mathcal{D} \\ Y_t \sim \pi_{\theta_{\text{old}}}(X)}} \left[\frac{1}{G} \sum_{\substack{q \leq Q \\ i \leq G}} c_q L_q^{(i)} \right] \quad (8)$$

where $\mathcal{D}$ denotes our input speech distribution and $\pi_{\theta_{\text{old}}}$ is a fixed version of the translation model that is replaced by π_{θ} every fixed number of updates τ .

4. Experiments

4.1. Architectural hyper-parameters

The backbone *Temporal* Transformer of Hibiki-Zero has a latent dimension of 2048 (8192 for the SiLU gating), 28 layers, 16 heads and local attention over 3000 tokens, i.e., 2B parameters and a 4min context. The *Depth* Transformer has a latent dimension of 1024 (4096 for the gating), 6 layers per codebook and 16 heads. It models $Q = 16$ audio codebooks for the output stream and the same for the input stream but only at training. We reduce the size of the model before RL by distillation into a smaller one using weight sharing among the codebooks of the *Depth* Transformer. Our final model architecture contains 3B parameters.

4.2. Training protocol

We train a multilingual-to-English speech translation system through the following steps, each with a cosine learning rate schedule and AdamW (Loshchilov & Hutter, 2019), with weight decay of 0.1, and momentum parameters (0.9, 0.95).

4.2.1. TEXT BACKBONE INITIALIZATION

We initialize the *Temporal* Transformer with Helium-1³ (Kyutai, 2025) weights, an open-source base text LLM with 2B parameters trained using filtered Common Crawl⁴ data.

4.2.2. AUDIO PRETRAINING

Starting from the pretrained text backbone, weights of a *Depth* Transformer modeling 8 audio codebooks are added to the architecture as well as audio tokens projection layers. We perform an audio pretraining with single stream audio as done by Labiausse et al. (2025) but on multilingual speech. Our data mixture comprises approximately 12% of audio in each input language, 50% of English and less than 2% of Italian. We train for 1M steps with a batch size of 144 and a learning rate of $2 \cdot 10^{-4}$. After this pretraining stage, we add new randomly initialized weights to the *Depth* Transformer corresponding to 8 additional audio codebooks so it can indeed model a total of $Q = 16$ audio codebooks in the next training stages. We then duplicate all the weights of the *Depth* Transformer to allow for future multistream training i.e. modeling $Q = 16$ audio codebooks for the input stream as well as the output stream. Appendix Figure 9 shows the cross-entropy of output text and audio tokens in pretraining.

4.2.3. COARSE SPEECH TRANSLATION TRAINING

We construct a large-scale multilingual-to-English speech translation dataset comprising 40,000 hours for each source language (French, Spanish, Portuguese, and German). Starting from a massive collection of multilingual audio, we extract 4 million single-speaker utterances, whose durations are between 30 and 75 seconds, and transcribe them using Whisper large-v3 (Radford et al., 2023). Transcripts are partitioned into sentences via Spacy’s `core_news_sm` and individually translated using MADLAD-3B (Kudugunta et al., 2023), after which we synthesize the target speech using the TTS system described in Section 3.2.2 with 10-second speaker conditioning. To ensure coarse translation alignments, we apply the silence insertion technique from Section 3.2.1 using $\delta = 0.5$ and $\mu = 2$. Scaling our training budget following Labiausse et al. (2025), we perform 500,000 gradient steps with a batch size of 96 and a learning rate of $3 \cdot 10^{-5}$, computing the loss on both source and target streams with source noise augmentation. Finally,

³huggingface.co/kyutai/helium-1-2b

⁴commoncrawl.org

Table 1. Objective comparison of Hibiki-Zero with Seamless (Barrault et al., 2023) and Hibiki (Labiausse et al., 2025) on short-form (Europarl-ST) and long-form (Audio-NTREX-4L) test data introduced in Section 4.3.

	SHORT-FORM						LONG-FORM					
	BLEU (↑)	ASR BLEU (↑)	ASR COMET (↑)	SPEAKER SIM. (↑)	END OFFSET (↓)	LAAL (↓)	BLEU (↑)	ASR BLEU (↑)	ASR COMET (↑)	SPEAKER SIM. (↑)	END OFFSET (↓)	LAAL (↓)
SEAMLESS												
FRENCH	33.8	32.8	76.6	19.1	2.4	2.8	27.8	23.9	33.7	44.4	3.2	6.2
SPANISH	34.4	33.6	79.1	21.9	2.6	2.7	29.9	25.2	36.1	42.6	2.8	6.5
PORTUGUESE	34.1	33.6	78.9	23.9	2.8	3.1	29.0	25.6	35.0	35.7	3.2	6.6
GERMAN	27.8	27.3	82.3	20.6	2.4	3.0	27.8	24.0	40.6	47.8	2.5	7.3
HIBIKI												
FRENCH	32.4	31.8	81.5	35.7	2.5	3.5	29.5	26.4	42.0	52.8	2.6	6.8
HIBIKI-ZERO												
FRENCH	35.0	34.6	80.3	49.5	2.1	2.8	30.6	28.7	43.7	61.3	2.3	6.1
SPANISH	33.8	33.9	80.3	57.0	2.3	3.1	32.3	31.5	42.3	64.6	2.6	5.6
PORTUGUESE	33.6	33.6	78.9	51.4	2.4	3.0	32.2	31.3	42.6	62.1	2.3	6.3
GERMAN	28.7	28.6	82.0	51.5	1.9	2.8	29.1	28.3	42.3	66.0	2.0	5.9

sequence termination is explicitly modeled by inserting a special input EOS token immediately following the source utterance and a separate EOS token in the text stream to indicate the end of generation. Appendix Figure 10 shows cross-entropy curves for output text and audio tokens during training, while Appendix Table 4 compares multilingual and monolingual model performance.

4.2.4. SPEECH TRANSLATION FINE-TUNING

We use the synthetic data generation method with natural pauses introduced in Section 3.2.2 to build a synthetic multilingual speech translation dataset of less than 200h in total. We fine-tune for 1K steps with a batch size of 16, a learning rate of $1 \cdot 10^{-6}$ and other configurations being similar to the previous phase described in 4.2.3. We then distill the model into a light copy of itself with codebooks weight sharing using the same dataset and 20K gradient updates.

4.2.5. REINFORCEMENT LEARNING

Starting from the light fine-tuned translation model, we use data from the speech translation training introduced in Section 4.2.3 and run our reinforcement learning process as described in Section 3.3. We train with a batch size of 32, a group size of 4, learning rate of $2 \cdot 10^{-7}$ and perform 2000 updates with $\tau = 20$. Sequences of length $T = 1500$ frames are generated using a temperature of 0.8 and top-k of 250 for both text and audio streams. Process rewards are computed every $n_w = 8$ input words and we set $\alpha = 0.4$ and $\epsilon = 0.2$. We use $c_0 = 100$ and $c_q = 1$ for $q \geq 1$ to balance loss between text and audio streams. The model is evaluated every $10 \cdot \tau$ updates on a valid set and we define Hibiki-Zero as the checkpoint with the best quality/latency trade-off according to objective evaluation metrics. Appendix Table 5 compares our base and fine-tuned models to Hibiki-Zero.

4.3. Evaluation datasets

Long-form data. We build Audio-NTREX-4L, a multilingual long-form ST dataset using text translations from the NTREX (Aepli et al., 2023) corpus. We select 300 examples for each source language and synthesize them using the following high-quality TTS from the industry: ElevenLabs⁵, Cartesia⁶ and Gradium⁷. We condition generations using voices from the multilingual dataset CML-TTS (de Oliveira et al., 2023). Audio-NTREX-4L contains around 15h of speech per TTS with an average duration of 45 seconds per sample and is split in balanced valid and test sets.

Short-form data. We filter data from Europarl-ST (Iranzo-Sánchez et al., 2020) and retain samples with realistic transcripts and duration between 2 and 20 seconds. We build valid and test sets, each with 1024 samples per source language for a total of 10h hours per set.

4.4. Evaluation metrics

Translation quality. We evaluate translation quality by transcribing generated speech using Whisper medium (Radford et al., 2023) and computing BLEU (Post, 2018) and COMET (Rei et al., 2020) scores with respect to a reference translation, referred to as ASR-BLEU and ASR-COMET. To reduce the impact of ASR errors, hypothesis and reference texts are normalized⁸ before computing BLEU scores. Since Seamless and Hibiki-Zero perform speech-to-text translation in parallel, we also compute BLEU and COMET scores using their text outputs. We use the XCOMET-XL model.⁹

Translation Latency. We rely on two common latency metrics known as End Offset and LAAL (Length-Adaptive

⁵elevenlabs.io/text-to-speech (“eleven-multilingual-v2 TTS”)

⁶cartesia.ai/sonic (“sonic-v2 TTS”)

⁷gradium.ai/#models (“default TTS”)

⁸github.com/openai/whisper/blob/main/whisper/normalizers

⁹github.com/Unbabel/COMET

Table 2. **Human evaluation.** Raters report Mean Opinion Scores (MOS) on a scale ranging from 0 to 100 for each audio sample.

INPUT LANGUAGE	MODEL	AUDIO QUALITY	SPEAKER SIMILARITY	SPEECH NATURALNESS
FRENCH	SEAMLESS	11.4 ± 3.1	21.1 ± 4.9	21.2 ± 3.8
	HIBIKI	62.9 ± 4.8	44.7 ± 5.1	57.0 ± 4.2
	HIBIKI-ZERO	64.5 ± 4.2	70.0 ± 5.1	67.2 ± 4.1
SPANISH	SEAMLESS	10.7 ± 2.6	21.2 ± 4.5	26.5 ± 4.4
	HIBIKI-ZERO	66.8 ± 3.9	69.0 ± 3.9	66.2 ± 4.9
PORTUGUESE	SEAMLESS	11.8 ± 3.1	32.5 ± 6.0	22.8 ± 3.9
	HIBIKI-ZERO	62.0 ± 4.1	60.7 ± 4.2	75.6 ± 3.4
GERMAN	SEAMLESS	15.6 ± 2.7	25.2 ± 4.9	26.4 ± 4.8
	HIBIKI-ZERO	73.5 ± 3.4	65.3 ± 4.3	69.9 ± 3.9

Table 3. Objective results of model adaptation to input Italian speech with 850 hours of finetuning data on short-form evaluation.

	BLEU (↑)	ASR BLEU (↑)	SPEAKER SIM. (↑)	END OFFSET (↓)	LAAL (↓)
SEAMLESS	32.5	32.0	22.2	3.0	3.5
OURS					
BASE	14.3	14.3	50.6	3.9	4.3
FINETUNED	31.4	31.0	55.2	3.7	4.5
FINETUNED + RL	32.1	31.9	54.2	3.0	3.5

Average Lagging). End Offset is defined as the time difference (in seconds) between the end of the last generated word and the end of the last word from the source. We compute LAAL following the method described by Papi et al. (2022) which defines it as an approximation of the average time (in seconds) between a source word and its translation. We use word-level emission timestamps $(d_i)_{1 \dots n_{\text{gen}}}$ produced by Whisper for n_{gen} words in the generated speech. We define $\gamma = \frac{\Delta_{\text{source}}}{\max(n_{\text{gen}}, n_{\text{ref}})}$ where Δ_{source} is the duration of the source speech and n_{ref} the number of words in the reference translation. The LAAL score is then computed as $\frac{1}{n_{\text{max}}} \sum_{i=1}^{n_{\text{max}}} d_i - (i-1)\gamma$ where $n_{\text{max}} = \min\{i | d_i \geq \Delta_{\text{source}}\}$.

Cross-lingual speaker similarity. For objective voice transfer evaluation, we use a standard model for speaker verification¹⁰ based on WavLM (Chen et al., 2022) and report the cosine similarity between the embeddings of the source and the generated speech.

Audio quality and naturalness. We rely on human raters to evaluate audio quality, speech naturalness and additional cross-lingual speaker similarity of generated audios. We conduct evaluations per input language using 50 samples and 20 raters for each model with 5 comparisons per rater.

¹⁰github.com/microsoft/UniSpeech (“WavLM Large”)

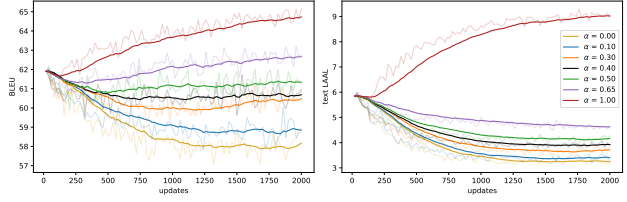


Figure 4. **Influence of hyperparameter α during RL.** We plot the BLEU score and text LAAL over training for various α (see Eq. (3)), starting from the same supervised model using $n_w = 8$.

4.5. Inference configuration

We encode audio with the streaming codec and feed the tokens to Hibiki-Zero while decoding the output tokens to obtain a streaming translation. At the end of the input, we force EOS tokens to our model input audio streams, and keep sampling until it produces its own text stream EOS. We use temperature of 0.8 and top-k of 250 for all tokens.

4.6. Results

Objective evaluations. Table 1 compares Hibiki-Zero against the best available baselines for simultaneous S2ST namely Seamless (Barrault et al., 2023) and Hibiki (Labi-aussie et al., 2025) with the latter only supporting French as input. Our model outperforms both baselines on long-form speech translation with more than 5pts of ASR BLEU, 20pts of speaker similarity and lower latency compared to Seamless. In the short-form setting, our approach outperforms Hibiki by 3pts of ASR BLEU while being faster and is on par with Seamless on the quality/latency trade-off but surpasses it on speaker similarity by more than 30pts.

Audio fidelity and speech expressivity. Human evaluations reported in Table 2 confirm the clear advantage of Hibiki-Zero compared to Seamless on speaker identity transfer but also show that it produces higher quality audio with better speech naturalness. Compared to Hibiki on a French-to-English task, our model reaches equivalent audio quality while being more natural with a better speaker similarity.

New language adaptation. Following our method from Section 4.2.3, we build a small coarse-aligned Italian-to-English ST dataset containing less than 1000 hours in each language. Starting from the base translation model obtained after the training stage described in Section 4.2.3, we fine-tune and apply our RL method for the Italian-to-English translation task only. Results are presented in Table 3 and show that we attain the same translation quality/latency trade-off as Seamless with better speaker similarity on an extension to Italian of our short-form evaluation data. As shown in Appendix Table 6, our model adapted to Italian also retains most of its capabilities on the original languages.

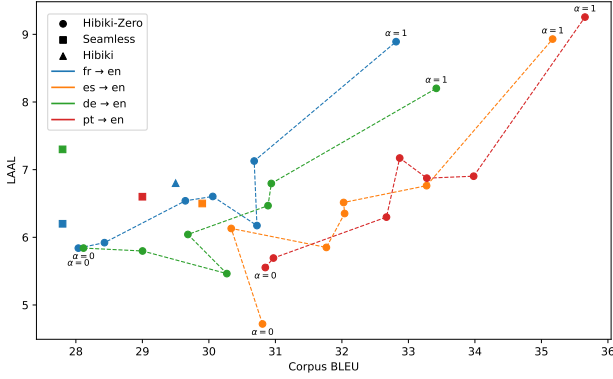


Figure 5. Comparison of translation quality/latency trade-off. We evaluate our approach by training with various values of α in $\{0.0, 0.1, 0.3, 0.4, 0.5, 0.65, 1.0\}$ and compare against baselines on the long-form evaluation benchmark Audio-NTREX-4L.

4.7. Ablations

We present ablation results in figures 4, 7 and 8 using exponential moving average smoothing for readability. Performance during RL is represented using BLEU and text LAAL metrics. They are computed every τ updates on a validation set using the output text stream of the model. As observed by Labiausse et al. (2025), we also notice very high BLEU scores (≈ 60) compared to evaluation scores (≈ 30). Indeed, our train and valid sets were obtained with the same data generation process described in Section 4.2.3 thus following the same translation style as MADLAD-3B (Kudugunta et al., 2023) that our ST models learn to replicate.

Ablation: Quality/Latency control during RL. We benchmark the effect of parameter α introduced in Section 3.3.1 which balances total and intermediate BLEU scores in the definition of process rewards. As illustrated in Figure 4, performing RL with high values of α leads to a higher translation latency but better overall translation quality as expected. On the contrary, lower values of α reduce latency further at the cost of a limited quality decrease. In Figure 5, we present quality/latency trade-off curves across languages pairs. Example translations for different values of α are provided in Appendix Figures 11, 12, 13, and 14.

Ablation: Process rewards computation frequency. Using $\alpha = 0.5$, we study the effect of parameter n_w introduced in Section 3.3.2 which controls how often process rewards are computed along a generated sequence. As shown in Figure 7, decreasing this parameter below 8 does not significantly impact the final quality/latency trade-off.

Ablation: Alternative configurations. In Figure 8, we compare alternative configurations that could be used for model development instead of our main setup referred to as *Reference experiment*. We keep $\alpha = 0.5$ and $n_w = 8$ fixed.

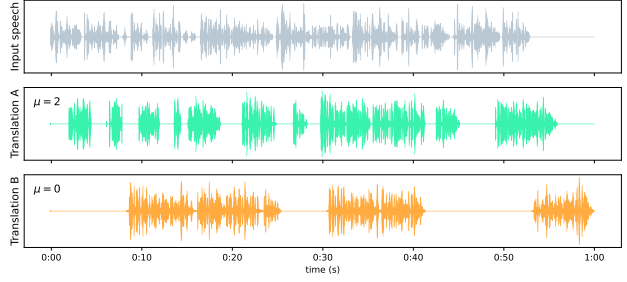


Figure 6. Illustration of coarse translation alignment patterns Waveform A is generated by a model trained on coarse alignments with random silences. Waveform B is generated by a model trained on coarse alignments with silences between sentences only.

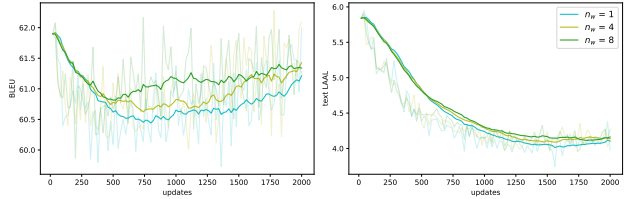


Figure 7. Influence of hyperparameter n_w during RL. We plot the BLEU score and text LAAL over training for various n_w (see Sec. 3.3.2) starting from the same supervised model using $\alpha = 0.5$.

Experiment (A) performs RL using the full translation of the reference input text instead of sentence-level prefixes to compute intermediate BLEU scores. This amounts to modify Equation 3 so it becomes $r_{\text{noprefix},t}^{(i)} = (1 - \alpha)\text{BLEU}(\hat{y}_t^{(i)}, y_T) + \alpha\text{BLEU}(\hat{y}_T^{(i)}, y_T)$. We observe better quality performance but at the cost of latency. According to us, this comes from intermediate BLEU scores being much noisier as translated references are too optimistic, making it harder to discriminate between sequences to optimize latency during RL.

Experiment (B) performs RL starting from a base model trained with full sentence delays between input and output speech meaning that $\delta_i = d_i$ for each sentence index i using notations from Section 3.2.1. Therefore, latency is much higher when starting RL and is reduced to around 6 seconds which remains far worse than the reference experiment. We also observe this behavior in preliminary experiments where RL was unable to teach the base model to start translation of an input sentence before it ends as the base model was never trained in that manner during supervised training. This justifies the use of $\delta_i < d_i$ when building coarse alignments so RL can benefit from exploration.

Experiment (C) performs RL starting from a base model trained with coarse alignments using sentence-level silences only ($\mu = 0$). We observe a degradation both in terms of

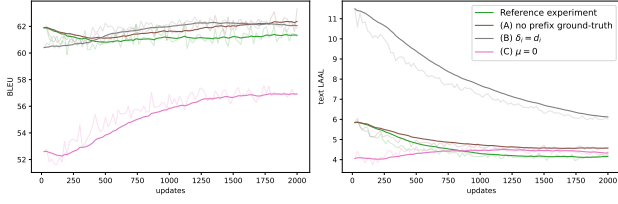


Figure 8. **Alternative configurations.** We use $\alpha = 0.5$ and $n_w = 8$ for all experiments. Experiment (A) uses the full translation of the input speech as reference to compute process rewards instead of sentence-level prefixes as in Equation 3. Experiment (B) performs RL on a supervised model trained with full sentence-delay i.e. $\delta_i = d_i$ for each input sentence of index i . Experiment (C) performs RL on a supervised model trained with coarse alignments using silences between sentences only i.e. $\mu = 0$.

quality and latency compared to the reference experiment. The loss of quality is expected when decreasing μ as we don't delay as much the output with respect to the input. The cause of higher latency is illustrated in Figure 6 where waveform B ($\mu = 0$) is a speech translation where silences are located between sentences only. This results in a higher average latency than waveform A ($\mu > 0$) which presents a better distribution of speech along time.

4.8. Limitations

This work proposes an efficient method to perform multilingual speech translation and shows promising results on new input language adaptation. However, while our model exhibits state-of-the-art speaker identity preservation, there is no way to control the intensity of the accent from the input language in the generated speech. Such control could be added by providing accent-annotated samples during supervised training and using conditioning at inference. Similarly, while the reinforcement learning stage enables steering the model toward a particular quality/latency trade-off, this trade-off cannot be adjusted at inference time.

5. Conclusion

We present Hibiki-Zero, a multilingual model for simultaneous and expressive speech and text translation without requiring word-level alignment of translation data for training. Our method leverages coarse sentence-level alignments to train a base model that is further refined through Reinforcement Learning using process rewards based on BLEU score only. Hibiki-Zero outperforms the state-of-the-art across multiple languages with better quality/latency trade-offs, speaker identity transfer and speech naturalness. Moreover, we demonstrate new language adaptation with our method using less than 1000 hours of speech data. We release Hibiki-Zero weights as well as our multilingual long-form evaluation dataset to benefit the research community.

Impact Statement

Our work aims at making simultaneous speech translation more accessible in multiple languages, by improving its quality but also by facilitating the research and development effort to build it. This technology has potential to improve human communication. While speech translation may impact interpreter employment in settings such as international meetings or live interviews, it is likely to have a net positive effect in contexts where interpreters are not typically available. We also acknowledge risks related to voice transfer but indicate that speech translation with voice transfer presents a lower spoofing risk than standard text-to-speech systems, as the output speech content is not directly controllable and the model's freedom to reformulate inputs is limited.

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Appendix

Table 4. Objective evaluations of multilingual and monolingual base supervised models. As the multilingual base model is trained on four times more data than each monolingual model, it has seen the same amount of each language after 400K updates as any monolingual model after 100K updates. For comparison, we provide evaluations after 100K and 400K updates for the multilingual model.

MODEL	SHORT-FORM					LONG-FORM				
	BLEU (↑)	ASR BLEU (↑)	SPEAKER SIM. (↑)	END OFFSET (↓)	LAAL (↓)	BLEU (↑)	ASR BLEU (↑)	SPEAKER SIM. (↑)	END OFFSET (↓)	LAAL (↓)
FRENCH										
BASE@100K	31.8	31.3	52.1	3.1	3.6	28.7	27.7	67.6	3.4	6.0
BASE@400K	34.5	34.1	53.1	3.0	3.7	31.1	28.8	67.4	3.4	6.6
BASE-FR@100K	34.2	33.4	52.5	3.0	3.6	29.8	28.6	67.3	3.2	6.1
BASE-ES@100K	5.3	5.1	39.6	4.4	5.1	5.9	5.5	60.1	7.7	11.2
BASE-PT@100K	6.7	6.6	41.2	4.6	5.0	7.9	8.0	62.0	5.8	8.5
BASE-DE@100K	1.6	1.6	41.6	4.3	5.6	2.0	1.7	64.1	8.3	10.8
SPANISH										
BASE@100K	31.5	31.4	59.2	3.4	4.1	31.1	30.9	69.5	3.9	6.6
BASE@400K	33.8	33.6	60.3	3.4	4.1	33.3	32.6	69.4	3.8	6.6
BASE-FR@100K	8.9	8.8	48.2	3.8	4.1	11.4	11.4	63.6	4.6	7.3
BASE-ES@100K	33.2	33.0	59.9	3.5	4.2	33.5	32.7	69.8	4.1	6.6
BASE-PT@100K	22.3	22.2	50.2	3.5	4.1	24.5	24.4	63.8	4.2	7.0
BASE-DE@100K	1.3	1.1	38.9	4.2	5.1	2.1	1.7	57.1	9.6	12.3
PORTUGUESE										
BASE@100K	31.7	31.4	53.5	3.7	4.2	31.7	30.7	66.9	3.3	6.6
BASE@400K	33.9	33.7	54.5	3.6	4.1	33.5	32.5	67.4	3.1	6.5
BASE-FR@100K	2.5	2.4	43.2	4.2	4.7	9.0	8.6	53.6	4.9	7.8
BASE-ES@100K	12.8	12.8	47.3	4.2	4.9	23.9	23.0	56.9	3.8	8.0
BASE-PT@100K	32.5	32.2	54.4	3.9	4.4	32.2	30.8	67.8	3.4	7.0
BASE-DE@100K	0.6	0.7	42.2	3.9	5.1	1.3	0.9	52.0	10.8	11.7
GERMAN										
BASE@100K	25.9	25.7	53.6	2.7	3.5	28.3	28.0	70.6	3.4	6.4
BASE@400K	28.3	28.0	54.6	2.7	3.6	31.1	30.5	70.5	3.3	5.8
BASE-FR@100K	0.7	0.7	35.6	4.4	4.6	1.4	1.2	54.5	12.1	10.0
BASE-ES@100K	0.9	0.8	35.4	4.6	5.2	1.8	1.1	50.9	14.5	16.0
BASE-PT@100K	0.7	0.7	33.3	4.1	4.6	1.6	1.4	50.2	8.1	9.6
BASE-DE@100K	28.6	28.3	55.6	2.9	3.8	30.8	29.9	71.4	3.4	6.2

Table 5. Objective evaluations of our base and fine-tuned models compared to Hibiki-Zero.

MODEL	SHORT-FORM						LONG-FORM					
	BLEU (↑)	ASR BLEU (↑)	ASR COMET (↑)	SPEAKER SIM. (↑)	END OFFSET (↓)	LAAL (↓)	BLEU (↑)	ASR BLEU (↑)	ASR COMET (↑)	SPEAKER SIM. (↑)	END OFFSET (↓)	LAAL (↓)
FRENCH												
BASE	34.4	34.0	79.2	53.1	3.0	3.6	31.1	29.7	43.1	67.5	3.5	6.2
FINETUNED	34.3	33.9	78.7	52.9	3.5	4.0	31.0	29.0	41.6	66.7	4.8	6.9
HIBIKI-ZERO	35.0	34.6	80.3	49.5	2.1	2.8	30.6	28.7	43.7	61.3	2.3	6.1
SPANISH												
BASE	34.0	33.7	80.1	60.2	3.3	4.1	33.8	32.8	45.3	69.8	3.8	6.3
FINETUNED	33.9	33.7	80.3	60.2	3.6	4.2	32.9	32.2	43.5	69.2	4.7	6.7
HIBIKI-ZERO	33.8	33.9	80.3	57.0	2.3	3.1	32.3	31.5	42.3	64.6	2.6	5.6
PORTUGUESE												
BASE	33.5	33.2	78.8	54.2	3.6	4.2	34.0	32.6	42.2	67.0	3.3	6.6
FINETUNED	34.0	33.9	78.8	55.5	3.9	4.3	33.7	31.7	42.2	67.5	4.6	7.3
HIBIKI-ZERO	33.6	33.6	78.9	51.4	2.4	3.0	33.2	31.3	42.6	62.1	2.3	6.3
GERMAN												
BASE	28.6	28.4	83.0	54.8	2.7	3.6	30.6	30.0	44.7	70.6	3.3	6.0
FINETUNED	28.1	27.9	82.4	54.6	2.9	3.7	30.6	29.4	44.8	70.3	4.5	7.0
HIBIKI-ZERO	28.7	28.6	82.0	51.5	1.9	2.8	29.1	28.3	42.3	66.0	2.0	5.9

Table 6. Comparison between Hibiki-Zero and our model adapted for Italian on original languages with long-form evaluation.

MODEL	BLEU (↑)	ASR BLEU (↑)	SPEAKER SIM. (↑)	END OFFSET (↓)	LAAL (↓)
FRENCH					
HIBIKI-ZERO	30.6	28.7	61.3	2.3	6.1
ITALIAN FINETUNED + RL	30.6	29.1	59.8	3.0	6.2
SPANISH					
HIBIKI-ZERO	32.3	31.5	64.6	2.6	5.6
ITALIAN FINETUNED + RL	31.1	30.3	62.8	2.6	6.3
PORTUGUESE					
HIBIKI-ZERO	33.2	31.3	62.1	2.3	6.3
ITALIAN FINETUNED + RL	32.9	31.3	56.4	2.7	6.5
GERMAN					
HIBIKI-ZERO	29.1	28.3	66.0	2.0	5.9
ITALIAN FINETUNED + RL	30.5	28.7	64.2	2.7	6.4

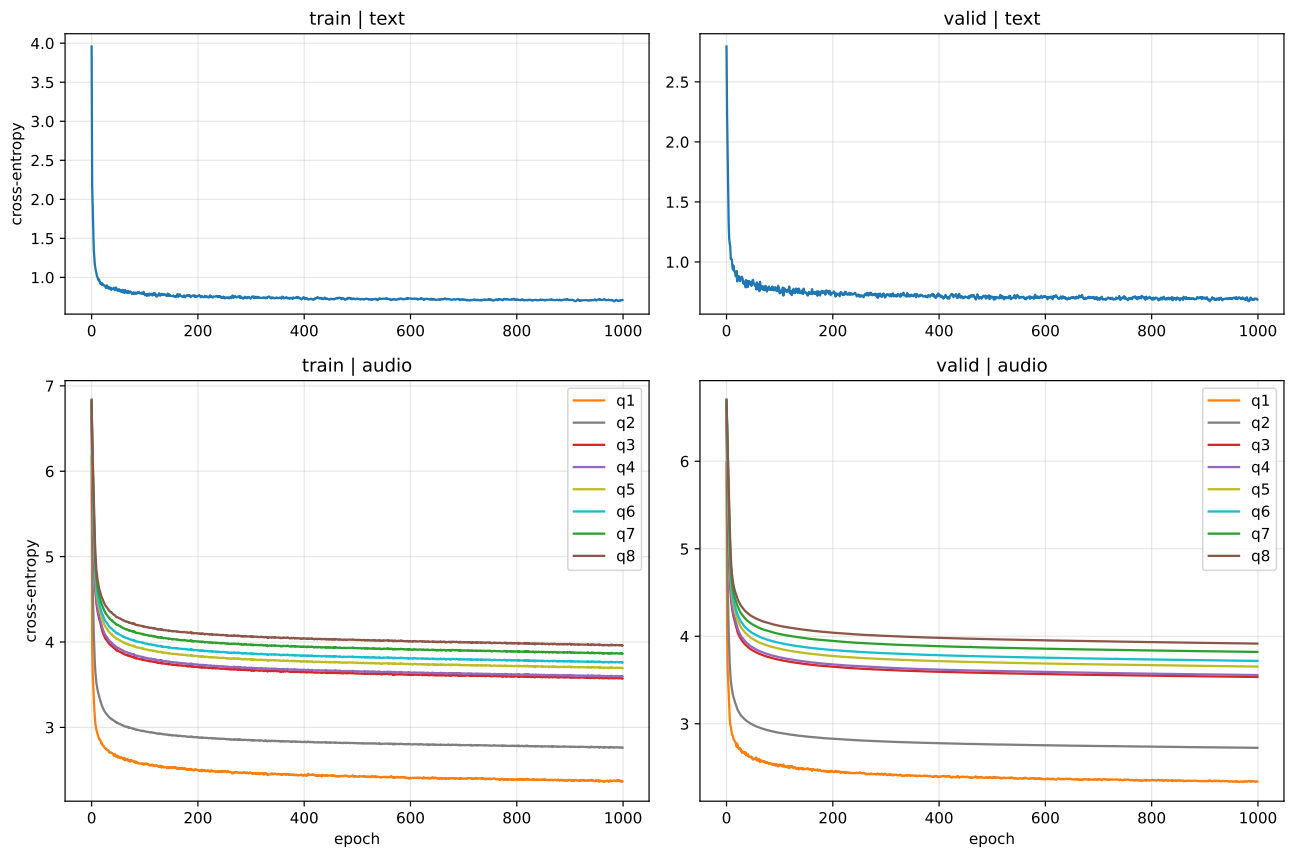


Figure 9. **Cross-entropy during audio pretraining.** Pretraining is performed with single stream audio using 8 layers of RVQ tokens.

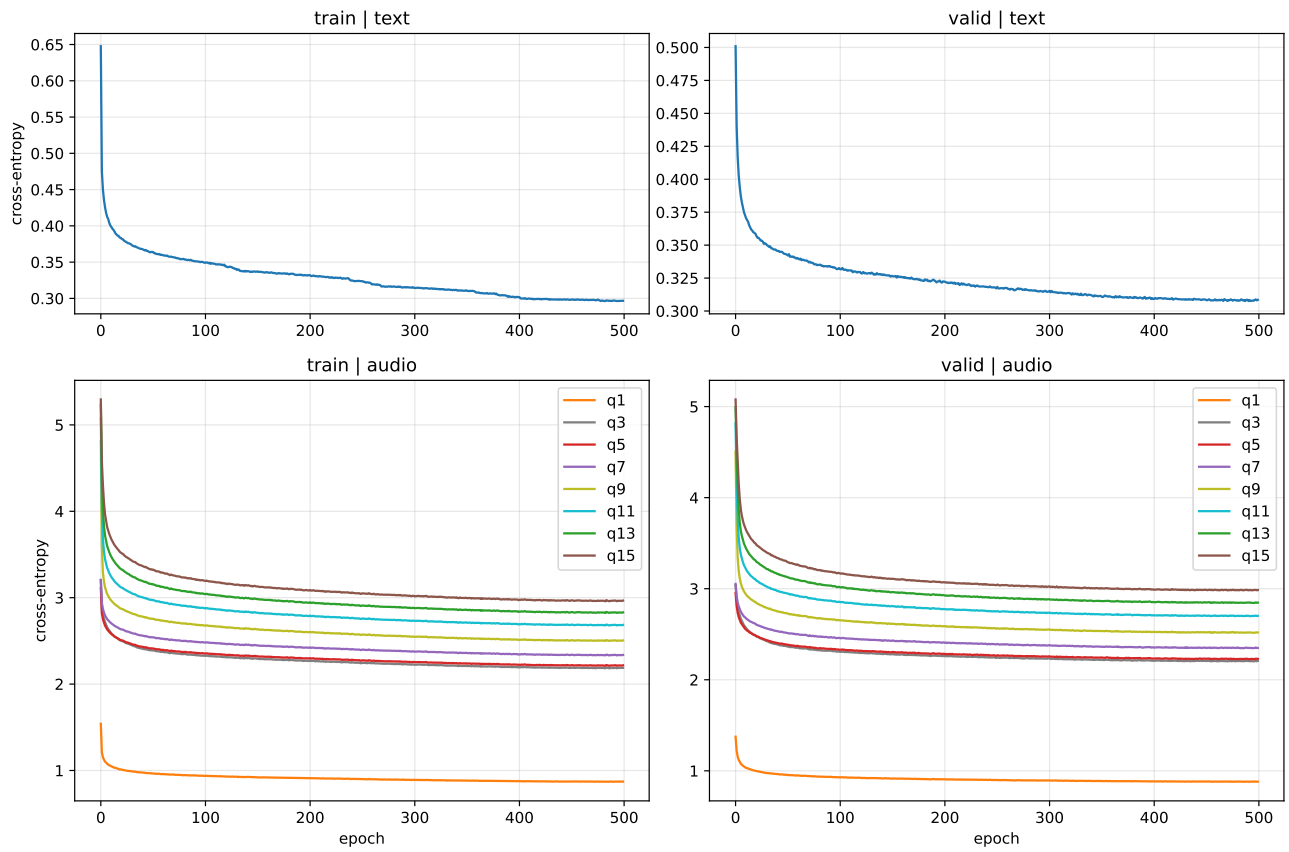


Figure 10. **Cross-entropy during coarse speech translation supervised training.** To improve readability, only the cross-entropy values of one out of every two layers of RVQ tokens from the model’s audio stream are shown.

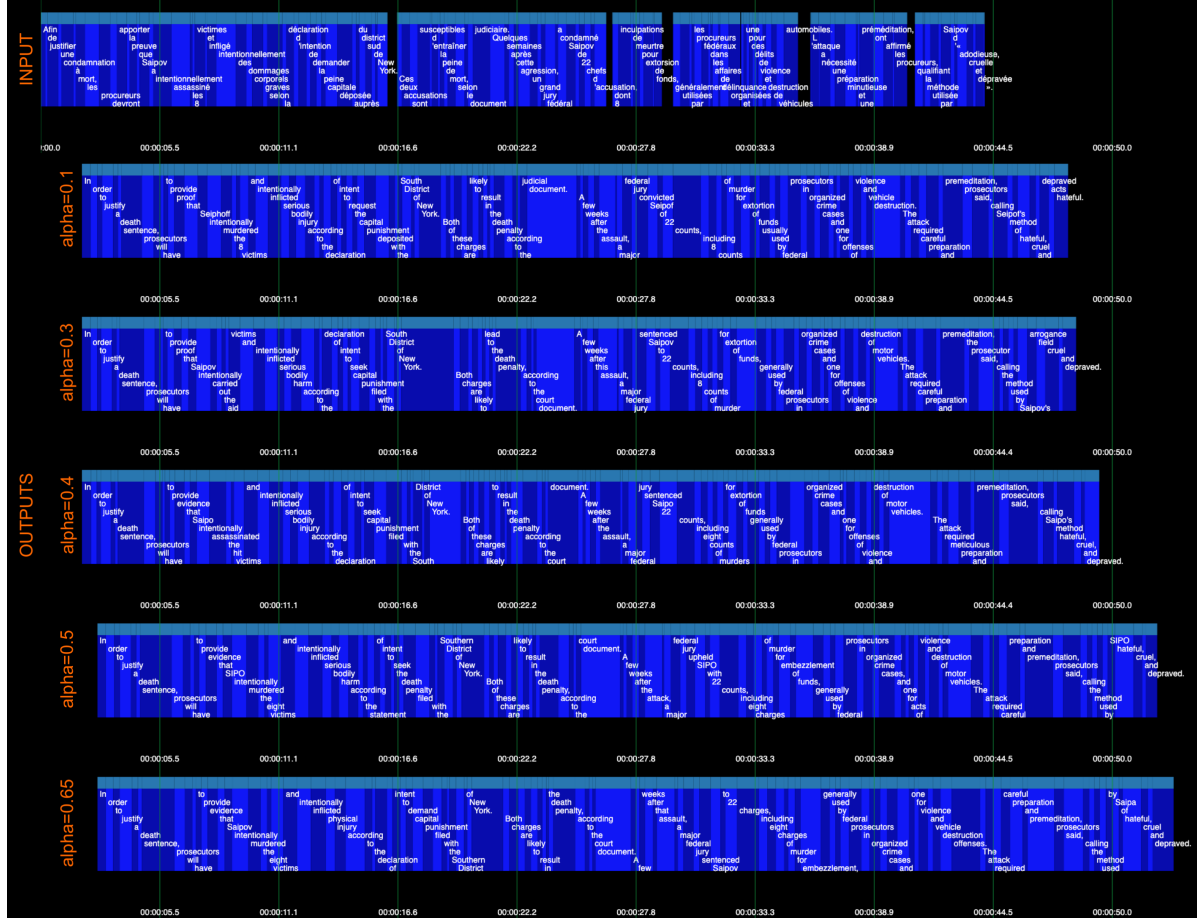


Figure 11. Example of outputs obtained from models trained with different α . The input is a French sample from Audio-NTREX-4L.

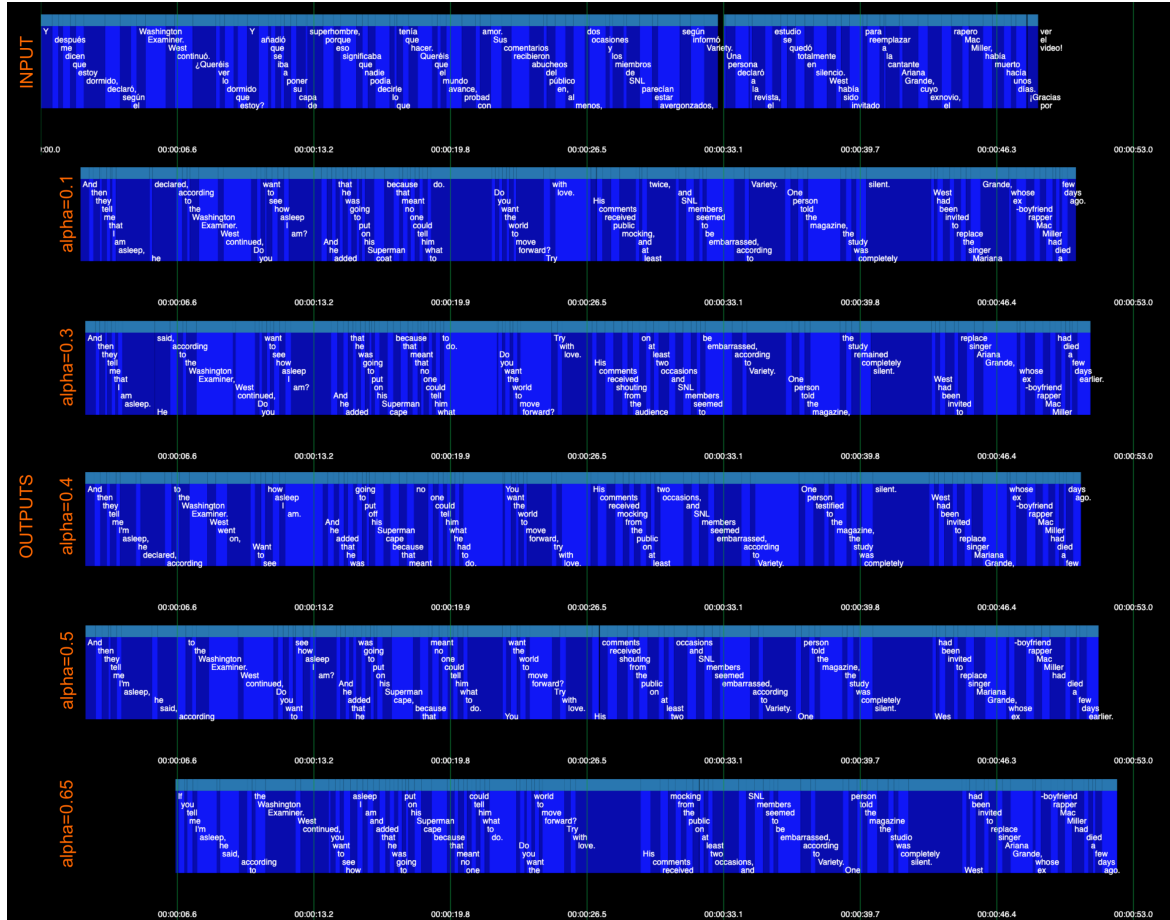


Figure 12. Example of outputs obtained from models trained with different α . The input is a Spanish sample from Audio-NTREX-4L.

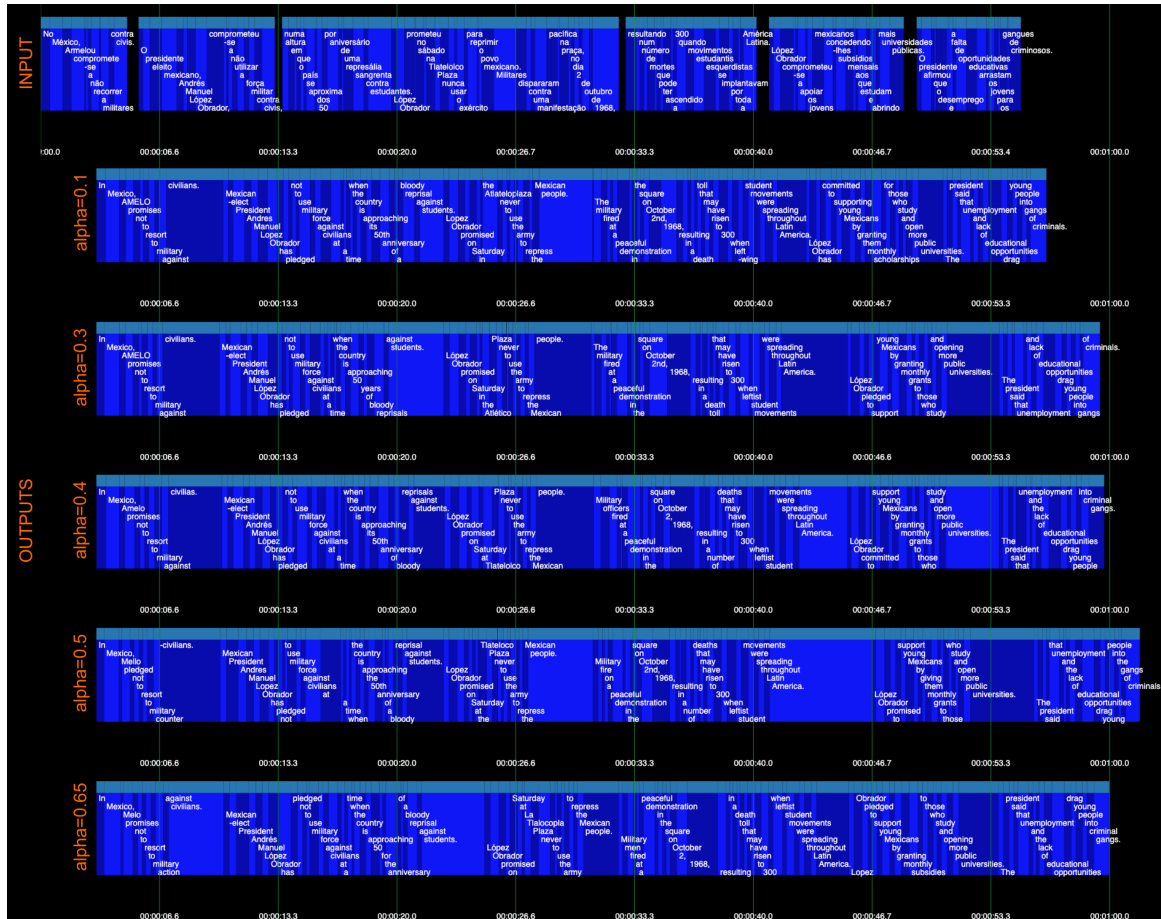


Figure 13. Example of outputs obtained from models trained with varo α . The input is a Portuguese sample from Audio-NTREX-4L.

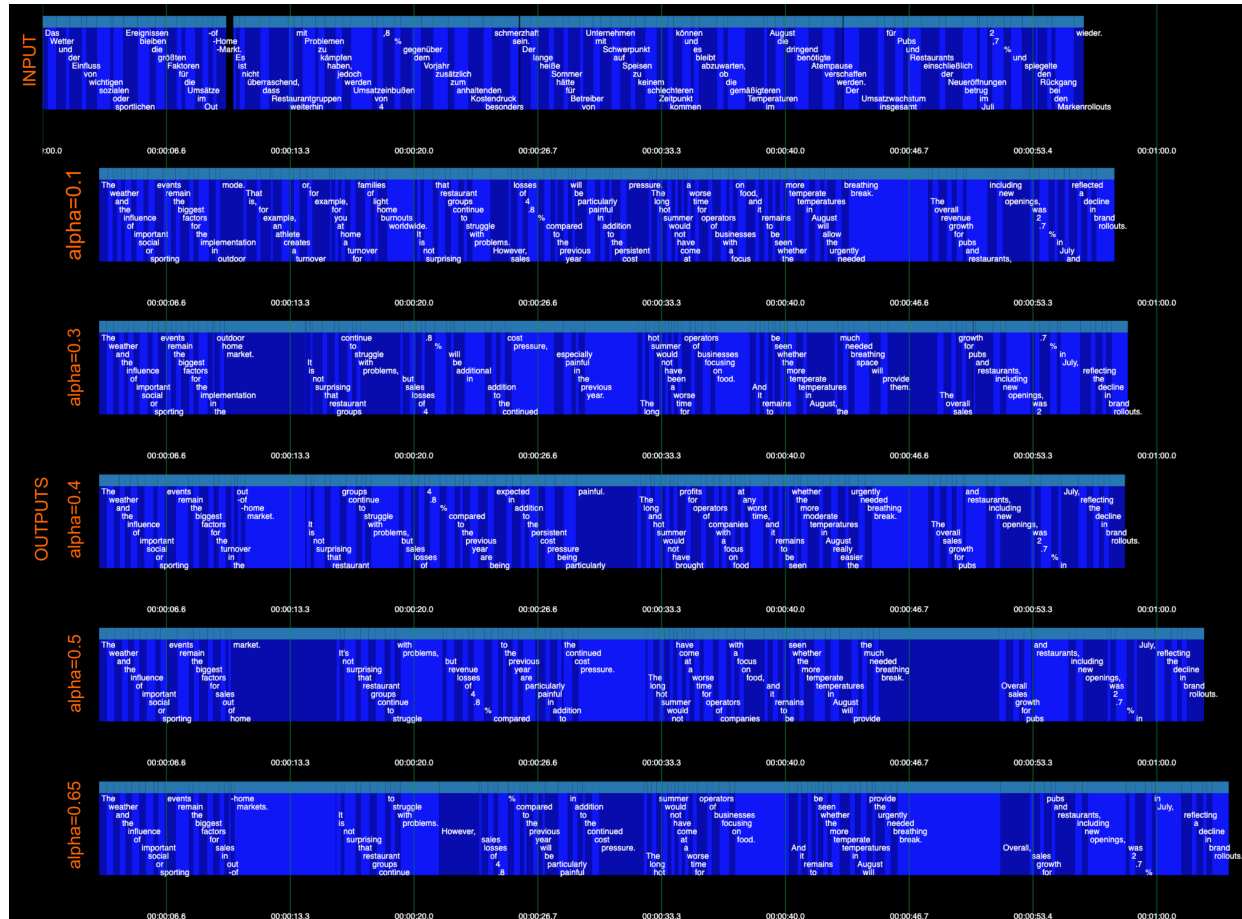


Figure 14. Example of outputs obtained from models trained with different α . The input is a German sample from Audio-NTREX-4L.